

6.1 Using the formula for time required to form a monolayer on the surface of a wafer,

$$t = \frac{N_s}{\Phi} = \frac{N_s \sqrt{2\pi m k T}}{P} = \frac{N_s \sqrt{M T}}{P \cdot 2.63 \times 10^{20}},$$

and substituting the molecular weight $M = 32$ for oxygen, as well as assuming approximately $N_s = 2.63 \times 10^{14}$ molecules per square centimeter on the surface, the time required to form a monolayer of oxygen on the surface of a wafer under the given conditions would be approximately

$$t = \frac{2.2 \times 10^{14} \sqrt{32 \times 300}}{101325 \times 2.63 \times 10^{20}} = 8.09 \times 10^{-10} \text{ s} = 0.817 \text{ ns}.$$

6.3 The impingement rate, Φ , given the provided parameters, would be

$$\Phi = \frac{2.63 \times 10^{20} \times 10^{-4}}{\sqrt{32 \times 300}} = 2.68 \times 10^{14} \frac{\text{molecules}}{\text{cm}^2 \cdot \text{s}}.$$

The mean free path, that is, the average distance traveled by a molecule before colliding with another molecule, is given by

$$\lambda = \frac{kT}{\sqrt{2}\pi P d^2},$$

where d is the diameter of the molecule ($7.2\text{\AA}$ for O_2). Solving for O_2 with the given parameters,

$$\lambda = \frac{1.38 \times 10^{-23} \times 300}{\sqrt{2} \times \pi 10^{-4} \times (7.2 \times 10^{-10})^2} = 17.98 \text{ m}.$$

Finally, converting pressure to torr,

$$\frac{10^{-4}}{133.3} = 7.5 \times 10^{-7} \text{ torr}.$$

6.5 Converting residual concentration to molecules per meter, $1000 \frac{\text{molecules}}{\text{cm}^2} = 10^7 \frac{\text{molecules}}{\text{m}^2}$. Then, finding pressure using the ideal gas law,

$$P = nkT = 10^7 \times 4.14 \times 10^{-21} = 4.14 \times 10^{-14} \text{ Pa}.$$

6.6 Assuming close packing of spheres, we may assume that a single layer of atoms will translate to

$$N_s = \frac{0.01^2}{4 \times 10^{-10}} = 6.25 \times 10^{14} \frac{\text{molecules}}{\text{m}^2} = 6.25 \times 10^{10} \frac{\text{molecules}}{\text{cm}^2}.$$

At a rate of 100 nanometers per minute, or 1.67 nanometers per second, the rate of formation $\Phi = 6.25 \times 10^{10} \times \frac{1.67 \times 10^{-9}}{4 \times 10^{-10}} = 26.1 \times 10^{10}$, we may solve for pressure as

$$26.1 \times 10^{10} = \frac{2.63 \times 10^{20} P}{\sqrt{M T}} \Rightarrow P = \frac{26.1 \times 10^{10} \times \sqrt{27 \times 300}}{2.63 \times 10^{20}} = 8.9 \times 10^{-8} \text{ Pa}$$

6.13 Evaporation is fast and low-cost, but produces weak film adhesion; sputtering produces a denser film with better adhesion, but is slower and more expensive; chemical vapor deposition produces an extremely conformal structure, but requires the highest temperatures of the three.